Lecture 3: Linear Regression

A model predicts y from x with y = w*x + b.

w is the weight (slope), b is the bias (intercept).

Example: w = 2, b = 1. For x = 3 the prediction is 2*3 + 1 = 7.

Loss: the mean squared error between predictions and labels.

MSE = (1/n) * sum_i (y_i - yhat_i)^2

Lecture 3: Gradient Descent

We update w by w <- w - lr * dL/dw.

lr is the learning rate. A small lr is slow; a large lr may diverge.

Example: w = 2, dL/dw = 4, lr = 0.1 gives w = 2 - 0.4 = 1.6.

Repeat until the loss stops decreasing.